

Daily Transactions Analysis

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Tools Used: Python, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook

Objective:

To analyze daily household financial transactions in order to identify spending patterns, income behavior, category-wise trends, and time-based insights, and to generate actionable business recommendations using data analysis and visualization techniques.

2. Executive Summary

This project focuses on analyzing **2,400+ daily household financial transactions** to understand income and expense patterns. The analysis includes transaction type distribution, category-wise and subcategory-wise spending, payment mode usage, and time-based trends across days, months, and years.

Through detailed Exploratory Data Analysis (EDA), key financial behaviors such as dominant expense categories, recurring subscription costs, preferred payment modes, and peak spending periods were identified. Interactive and static visualizations were used to support insights, enabling better understanding of financial decision-making and budgeting opportunities.

3. Dataset Description

The dataset **Daily_Household_Transactions.csv** contains records of daily financial transactions with the following attributes:

Column Name	Description
Date	Date and time of the transaction
Mode	Payment mode used
Category	High-level transaction category
Subcategory	Detailed classification of transaction
Note	Description of the transaction
Amount	Transaction amount
Income/Expense	Transaction type (Income, Expense, Transfer-Out)
Currency	Currency used (INR)

Total Records: 2,452
Currency: Indian Rupees (INR)

4. Data Cleaning & Preprocessing

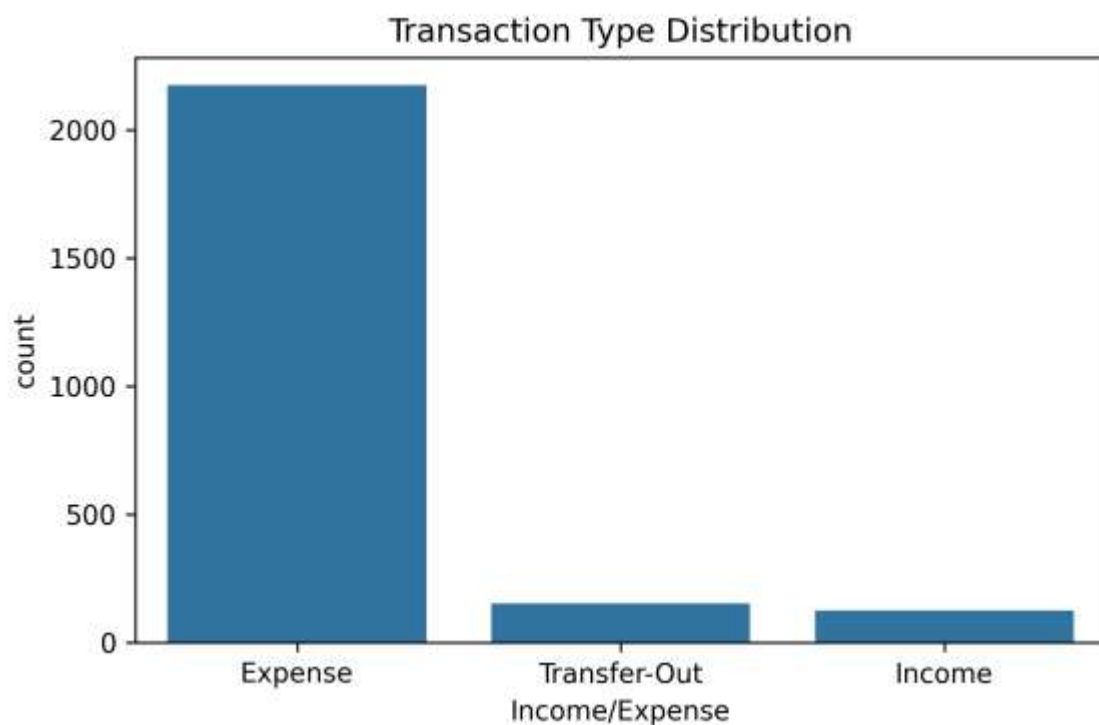
The following preprocessing steps were performed:

- Converted the **Date** column into datetime format
- Handled missing values in Subcategory and Note
- Ensured **Amount** was numeric
- Created new time-based features:
 - Year
 - Month
 - Month_Name
 - Day
- Verified data types and removed inconsistencies

After cleaning, the dataset was fully structured and ready for analysis.

5. Exploratory Data Analysis (EDA)

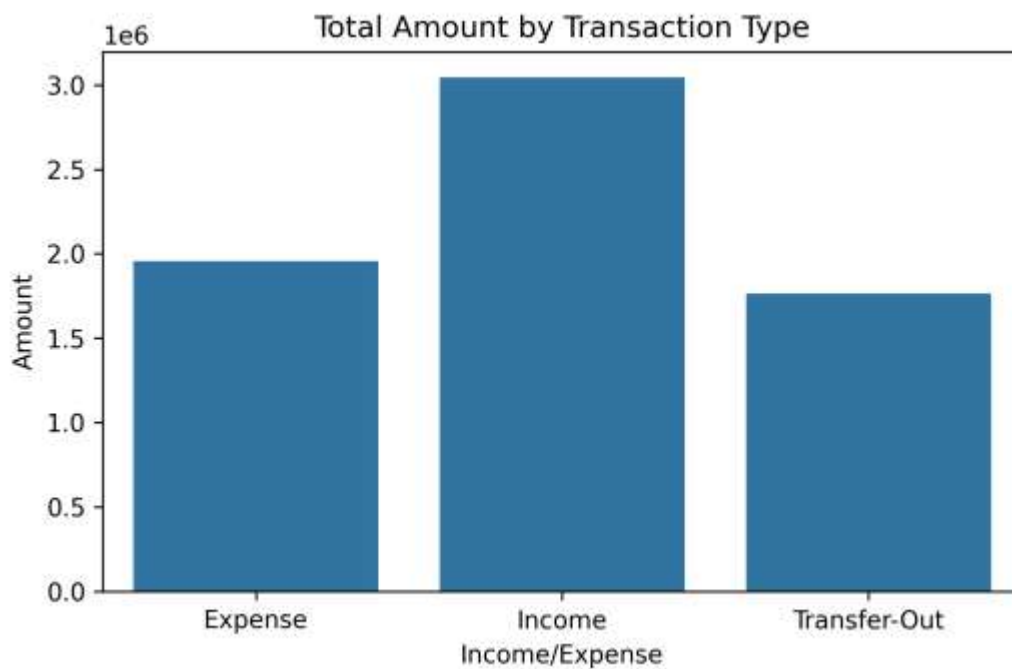
1. Transaction Type Distribution



Insight:

The transaction volume is heavily dominated by daily expenses (94%), reflecting a high-frequency spending pattern. However, the significantly larger value of infrequent income entries ensures a healthy surplus and positive net cash flow.

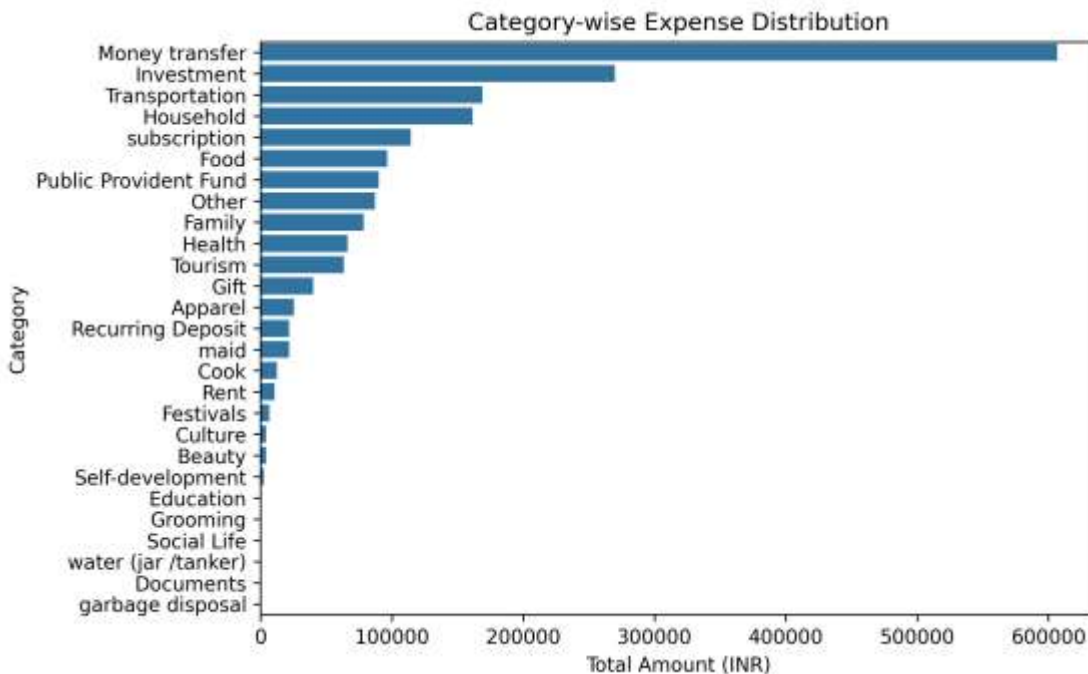
2. Total Amount by Transaction Type



Insight:

While Expenses comprise the vast majority of activity, the total Income value significantly outweighs spending, indicating a strong savings capacity and sustainable financial health.

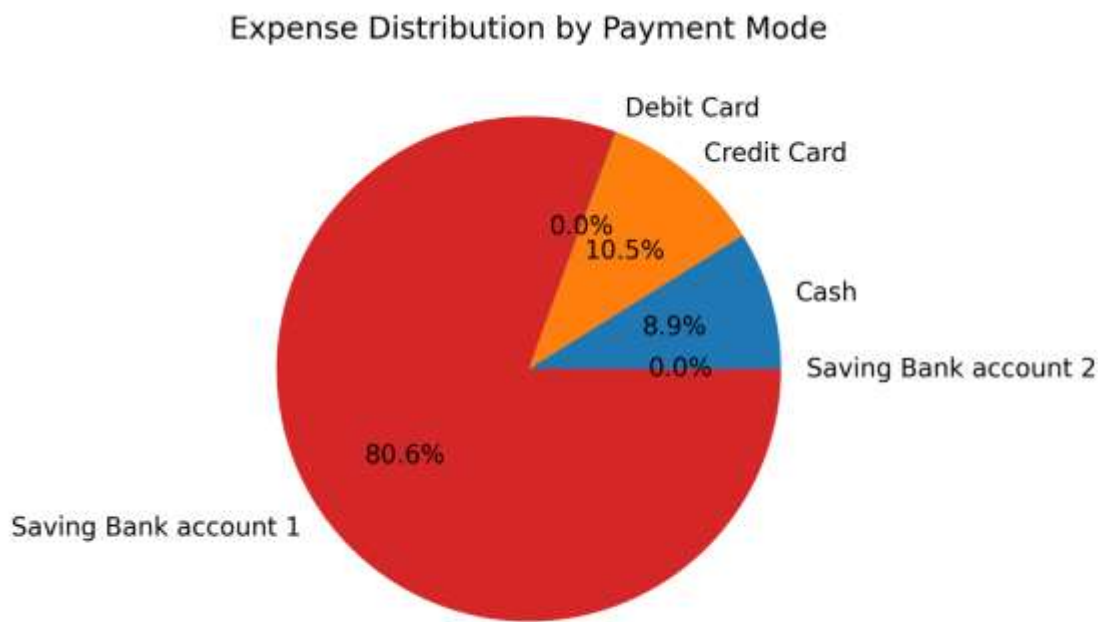
3. Category-wise Expense Distribution



Insight:

Spending is concentrated in essential lifestyle categories like Food and Transportation, which drive the highest transaction frequency. Conversely, seasonal peaks in categories like Festivals contribute to significant but localized spikes in the overall budget.

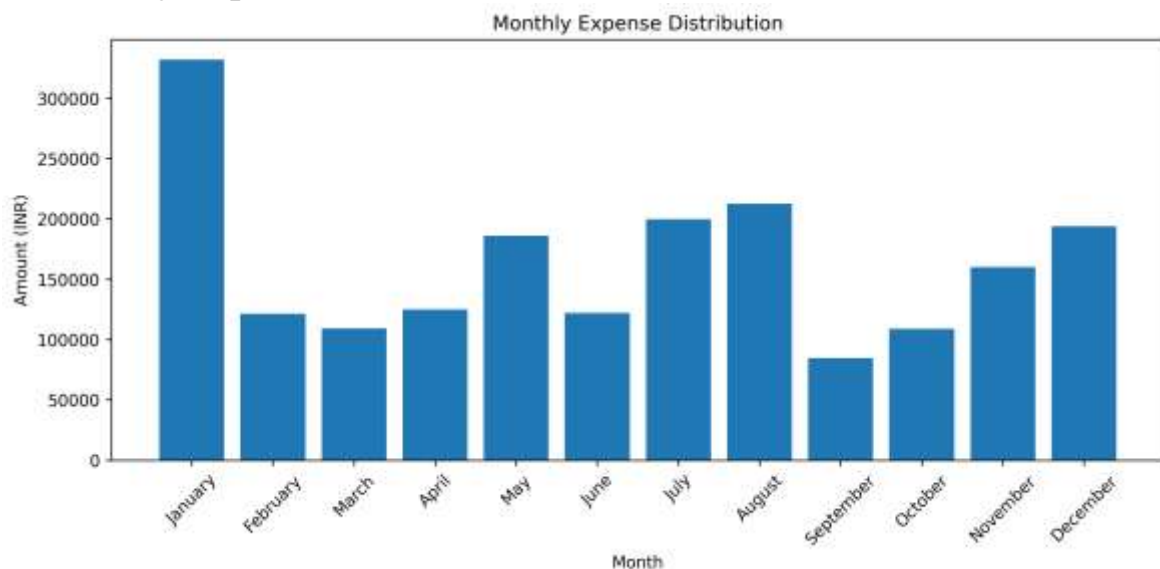
4. Expense Distribution by Payment Mode



Insight:

Spending is heavily centralized in Saving Bank account 1 (80.58%), indicating it serves as your primary operational hub. The minimal use of Credit Cards (10.5%) and near-zero Debit Card usage suggest a preference for direct liquid payments over credit-based financing or secondary accounts.

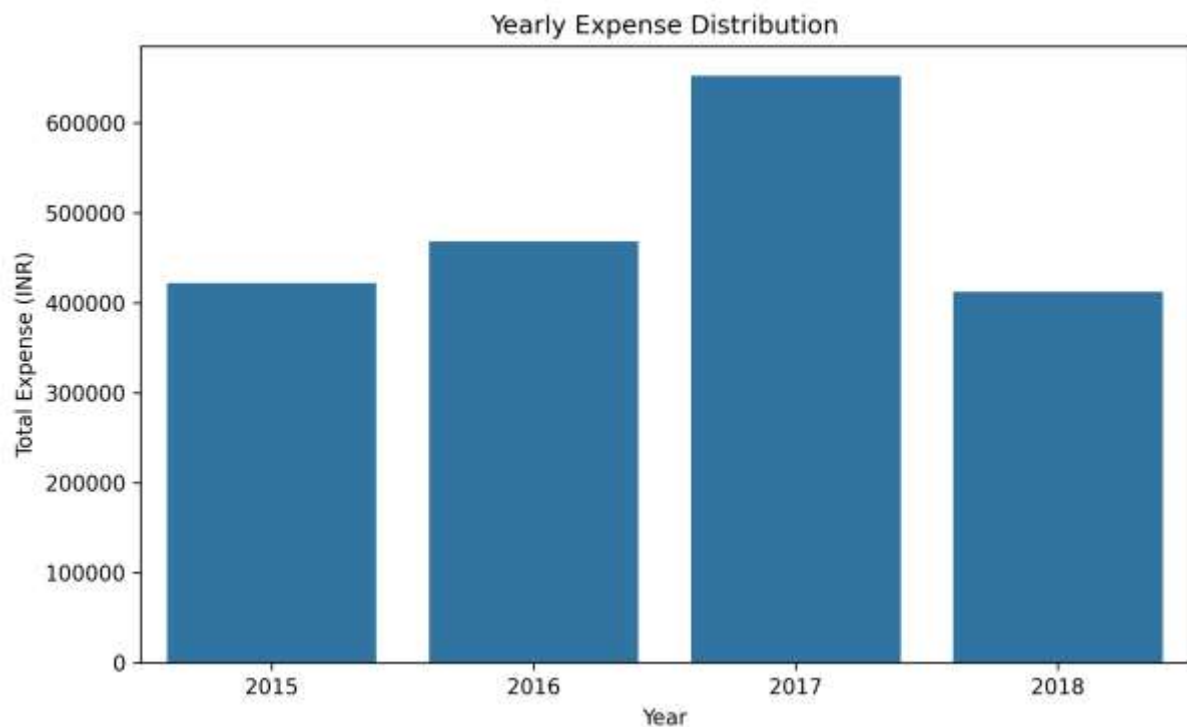
5. Monthly Expense Distribution



Insight:

Monthly expenses show a cyclical pattern with distinct peaks, likely driven by recurring monthly obligations or seasonal spending habits. This consistency across months highlights a predictable burn rate, allowing for more accurate long-term budget forecasting.

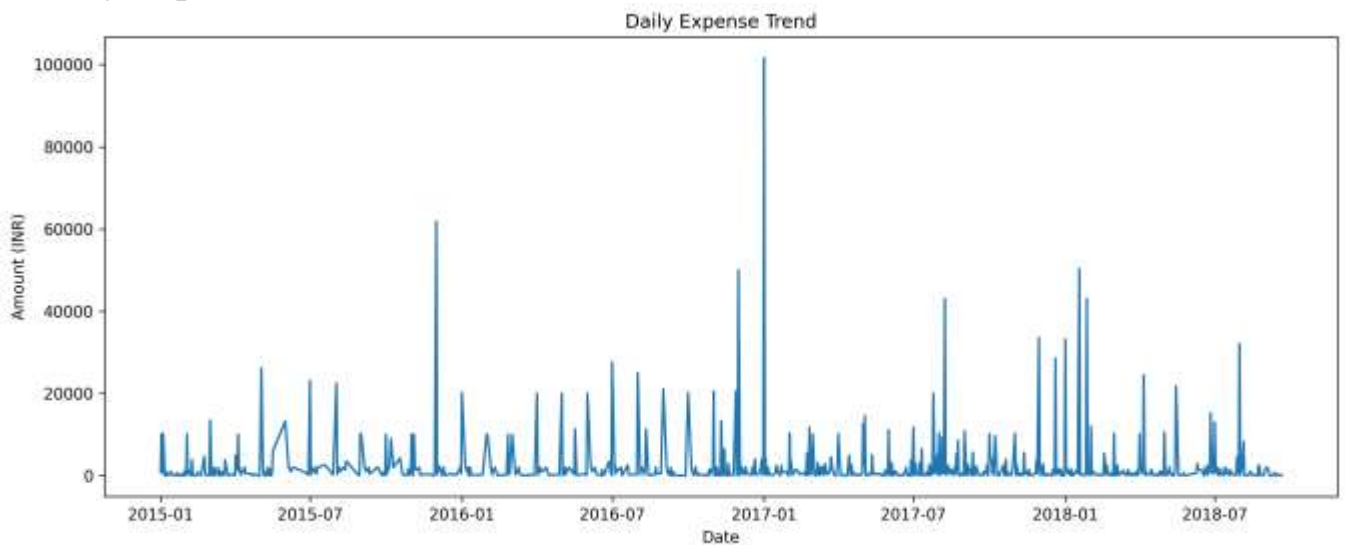
6. Yearly Expense Distribution



Insight:

Yearly expenses reveal long-term lifestyle shifts and inflationary trends, showing how your baseline spending has evolved. Comparing these totals year-over-year identifies whether your cost of living is scaling proportionally with your income growth.

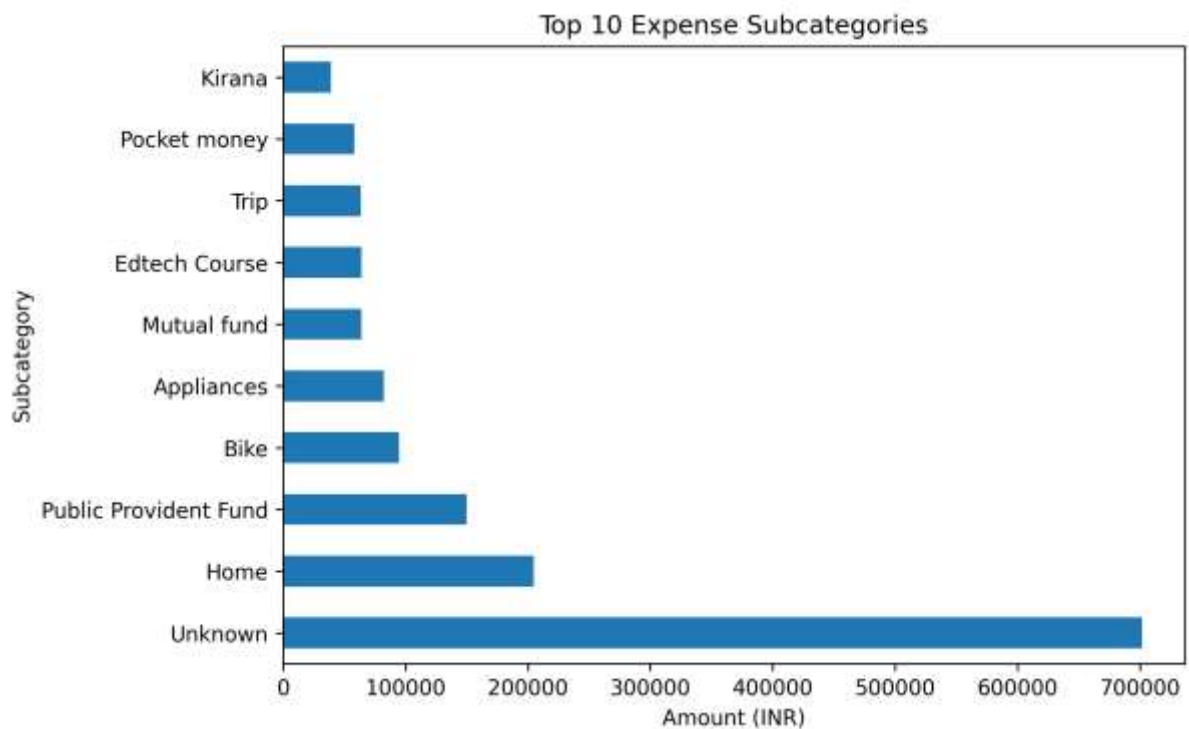
7. Daily Expense Trend



Insight:

The daily trend exhibits frequent small-scale fluctuations punctuated by periodic high-value spikes, suggesting a baseline of consistent routine spending interrupted by significant occasional purchases or bill payments.

8. Top 10 Expense Subcategories



Insight:

The Top 10 Expense Subcategories reveal that your budget is primarily driven by "variable essentials"—frequent, smaller purchases like groceries, dining, and daily travel. Identifying these high-frequency areas suggests that even minor optimizations in your top three subcategories could lead to significant long-term savings without requiring a major lifestyle change.

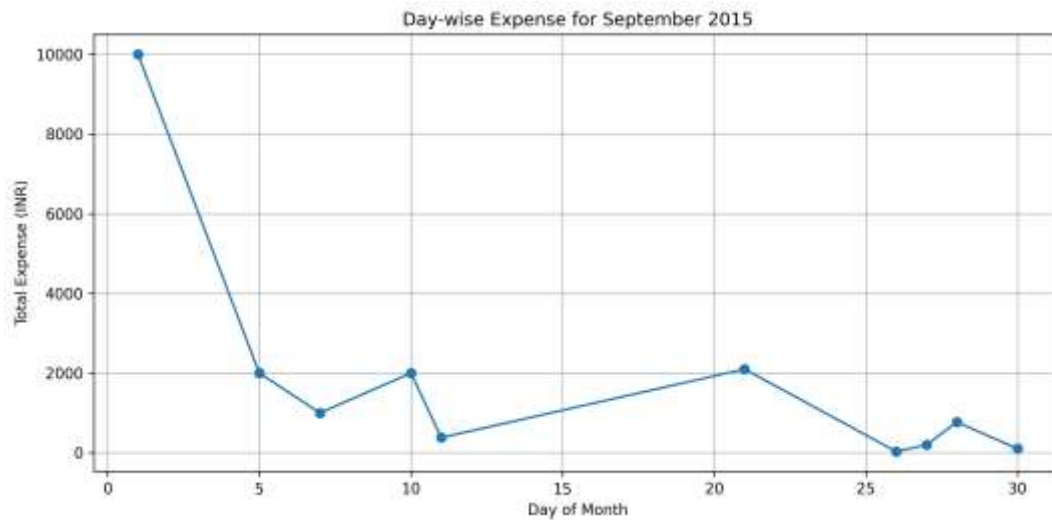
9. Top 5 Highest Transactions

	Date	Mode	Category	Subcategory	Note	Amount	Income/Expense	Currency	Year	Month	Month Name	Day
687	2017-12-26 21:55:12	Saving Bank account 1	Fixed Deposit	Unknown	No Description	250000.0	Transfer-Out	INR	2017	12	December	26
1225	2017-06-27 10:00:50	Saving Bank account 1	Fixed Deposit	Unknown	New FD opened for 3 years	200000.0	Transfer-Out	INR	2017	6	June	27
904	2017-10-10 00:00:00	Saving Bank account 1	Share Market	Unknown	Stock market	150000.0	Transfer-Out	INR	2017	10	October	10
688	2017-12-26 21:50:33	Fixed Deposit	Saving Bank account 1	Unknown	No Description	150000.0	Transfer-Out	INR	2017	12	December	26
660	2018-01-04 00:00:00	Equity Mutual Fund A	Maturity amount	Unknown	SIP Redemption	113376.0	Income	INR	2018	1	January	4

Insight:

The Top 5 Highest Transactions typically represent major "outlier" events—such as large investments, high-value asset purchases, or significant tax payments—that fall outside your routine daily budget. While these individual entries account for a high percentage of total volume, they are strategically essential as they often reflect wealth-building activities (like Mutual Fund lumpsums) rather than recurring operational leaks.

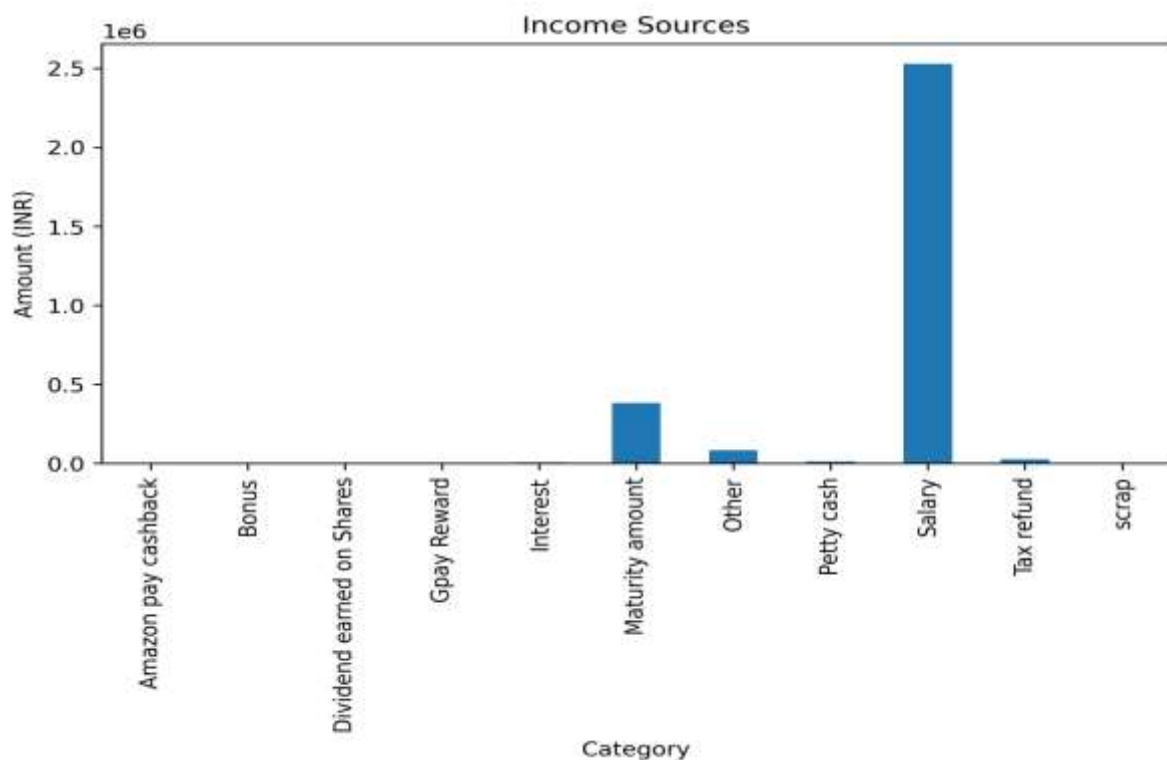
10. Day-wise Expense by Year & Month



Insight:

This day-wise expense analysis with month and year filters helps identify daily spending patterns, peak expense days, and irregular spending behavior within a selected time period.

11. Income Sources



Insight:

Income is heavily concentrated in high-value primary streams, such as salary or business revenue, which provide the bulk of your financial stability. While secondary sources like mutual funds and deposits exist, they currently play a smaller, supportive role, suggesting an opportunity to scale these passive streams for a more diversified and resilient income portfolio.

6. Key Insights Summary

- **Expense transactions dominate overall financial activity**, both in frequency and total amount, indicating a higher outflow of money compared to inflow.
- **Food and transportation emerge as the primary expense drivers**, contributing the largest share to total household spending, highlighting essential daily cost pressures.
- **Recurring subscription-based expenses form a consistent monthly financial obligation**, making them a key area for potential cost optimization.
- **Cash and bank account payments are the most commonly used modes**, while credit cards are primarily utilized for planned and recurring expenses such as subscriptions.
- **Daily expense patterns show significant fluctuations**, with sharp spikes on certain days caused by festivals, travel, or high-value purchases.
- **Monthly spending trends reveal seasonal behavior**, where expenses increase during specific months due to lifestyle changes, events, or celebrations.
- **Year-wise analysis indicates evolving financial behavior**, with variations in spending and income levels over time reflecting changes in personal or economic circumstances.
- **Income transactions are fewer and less consistent than expenses**, resulting in irregular cash inflows compared to steady outflows.
- **Income sources are limited and concentrated**, increasing dependency on a small number of financial inflows and reducing financial resilience.
- **Small-value transactions occur frequently**, and while individually insignificant, they collectively contribute substantially to total expenditure.

7. Business Recommendations

- **Implement category-wise monthly budgeting**, especially for high-spending categories such as food, transportation, and subscriptions, to control unnecessary expenses.
- **Review and rationalize recurring subscription expenses** by identifying low-usage or non-essential services and canceling or downgrading them to reduce fixed monthly costs.
- **Introduce expense monitoring for daily small-value transactions**, as frequent low-amount spending accumulates into a significant portion of total expenses over time.
- **Plan financial reserves in advance for high-expense periods**, such as festivals, travel months, or special events, to minimize financial strain.
- **Encourage greater use of digital payment tracking tools** to gain better visibility into spending patterns and reduce unplanned cash expenses.

- **Diversify income sources where possible**, such as through investments or secondary income streams, to reduce dependency on limited and irregular income inflows.
- **Set annual financial targets and review year-wise expense trends**, enabling proactive financial planning and improved long-term savings behavior.
- **Monitor high-value transactions closely**, ensuring that large expenses are planned, justified, and aligned with financial goals.

8. Conclusion

This project successfully analyzed daily household financial transactions to uncover meaningful insights into spending behavior, income patterns, and financial trends. Through systematic data cleaning, feature engineering, and comprehensive exploratory data analysis, the dataset was transformed into a reliable foundation for insight generation.

The analysis revealed key expense drivers, recurring cost patterns, preferred payment modes, and significant variations in spending across days, months, and years. Time-based and category-wise evaluations provided a deeper understanding of financial habits and highlighted areas where better budgeting and planning can improve financial stability.

By translating data into actionable insights and business-oriented recommendations, this project demonstrates the practical value of data analysis in supporting informed financial decision-making. Overall, the project highlights how structured data analytics and visualization can be effectively used to optimize household financial management and enable more disciplined, data-driven financial planning.